

# Yen-Hsiang (Robert) Huang

✉ yhhuang@gatech.edu 📞 +1(470)923-6469

## Educations

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| <b>Ph.D. in Electrical and Computer Engineering   GPA: 4.0/4.0</b><br>Georgia Institute of Technology (Gatech), Atlanta, GA, USA | Sept. 2022 - Present   |
| <b>B.S. in Electrical Engineering   GPA: 4.06/4.30</b><br>National Taiwan University (NTU), Taipei, Taiwan                       | Sept. 2017 - July 2022 |

## Research Experience

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| <b>Graduate Research Assistant</b><br>SCCAD (former GTCAD), Georgia Institute of Technology, Advisor: Prof. Sung Kyu Lim                                                                                                                                                                                                     | Sept. 2022 - Present    |
| <b>3D IC Optimization through 3D-VIA Reduction &amp; Inter-Die Cell Relocation</b> <ul style="list-style-type: none"><li>Timing optimization through reduced critical path snaking and inter-die cell relocation.</li></ul>                                                                                                  | Fall 2024 - Present     |
| <b>Semiconductor Research Corporation (SRC) – JUMP 2.0 CHIMES</b> <ul style="list-style-type: none"><li>Bridge the PPA gap between legacy node 3D ICs and advanced nodes 2D ICs</li><li>Reduced WNS of legacy node 3D ICs by 50-93% compared to SOTA, achieving 87.9-97.7% of advanced node 2D performance</li></ul>         | Fall 2023 - Spring 2024 |
| <b>Die Bonding Bumps and Pads legalization for 3D ICs</b> <ul style="list-style-type: none"><li>Developed a 3D via legalization stage with a refinement step to legalize 3D vias with minimize runtime impact</li><li>Our flow is capable of eliminating all 3D via overlaps when via utilization is less than 40%</li></ul> | Sept. 2022 - May 2023   |
| .....                                                                                                                                                                                                                                                                                                                        |                         |
| <b>Individual Study</b><br>EDA Lab., National Taiwan University, Advisor: Prof. Yao-Wen Chang                                                                                                                                                                                                                                | Apr. 2020 - Sept. 2021  |
| <b>Independent Study on Physical Design</b> <ul style="list-style-type: none"><li>Participated in physical-design-related global contests three times in different teams, and get into the top 5 two times (ISPD contest 2021, ICCAD contest 2021)</li></ul>                                                                 |                         |
| ALCom Lab., National Taiwan University, Advisor: Prof. Jie-Hong Jiang                                                                                                                                                                                                                                                        | Sept. 2020 - June 2021  |
| <b>Independent Study on Reversible circuit Synthesis</b> <ul style="list-style-type: none"><li>Implemented a reversible circuit synthesizer for controlled-NOT (CNOT) gates</li></ul>                                                                                                                                        |                         |

## Work Experience

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| <b>R&amp;D Engineer</b><br>Synopsys Inc.                                                                                                                                                                                                                                                                                                       | <b>Synopsys Headquarters</b><br>May 2025 - Dec. 2025 |
| <ul style="list-style-type: none"><li>Investigated AI-based signal-integrity (SI) analysis for 2.5D ICs, using graph neural networks (GNNs) to predict final-stage SI performance from early-stage design data.</li><li>Independently implemented two provided prototypes in C++ and merged the resulting code into the main branch.</li></ul> |                                                      |
| <b>Software Engineer R&amp;D Intern</b><br>Synopsys Inc. - Concurrent Clock-and-Data (CCD) Team                                                                                                                                                                                                                                                | <b>Hsinchu, Taiwan</b><br>July 2021 - Aug. 2021      |
| <ul style="list-style-type: none"><li>Enhanced the flip-flop merging algorithm in Synopsys IC Compiler II (ICC II) by considering the flip-flop's timing (slack)</li><li>Improved the timing quality in 13 real industrial benchmarks: 5.14%/3.47% in worst/total negative slack</li></ul>                                                     |                                                      |
| .....                                                                                                                                                                                                                                                                                                                                          |                                                      |
| <b>Software Engineer R&amp;D Intern</b><br>Cadence Inc. - Conformal Team                                                                                                                                                                                                                                                                       | <b>Hsinchu, Taiwan</b><br>June 2020 - Sept. 2020     |
| <ul style="list-style-type: none"><li>Provided electronic change order (ECO) test cases in a 5-person team for Cadence Conformal's robustness test</li><li>Generated 2000+, 5 kinds of cases in which Conformal acts unusually and proposed 3 improved flows to fix</li></ul>                                                                  |                                                      |

## Publication

- Yen-Hsiang Huang** and Sung Kyu Lim, "CLK-3D: RC-Calibrated Differentiable Clock-Tree Optimization for 3D ICs," IEEE/ACM International Conference on Computer-Aided Design, 2026.
- Yen-Hsiang Huang** and Sung Kyu Lim, "Snake-3D: Differentiable Learning for Cross-Tier Logic Path Snaking Optimization in 3D ICs," IEEE/ACM International Conference on Computer-Aided Design (ICCAD), 2025. [PDF]
- Yen-Hsiang Huang**, Sai Pentapati, Anthony Agnesina, Moritz Brunion, and Sung Kyu Lim, "On Legalization of Die Bonding Bumps and Pads for 3D ICs," IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2024.
- Sai Pentapati, Anthony Agnesina, Moritz Brunion, **Yen-Hsiang Huang**, and Sung Kyu Lim, "On Legalization of Die Bonding Bumps and Pads for 3D ICs", ACM International Symposium on Physical Design (ISPD), 2023.

## Honors and Awards

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| <b>Honorable Mention Award in the ACM ISPD contest 2021</b> <ul style="list-style-type: none"><li>4th place worldwide in the contest.</li></ul>                                | March 2021             |
| <b>TSMC scholarship for undergraduate student</b> <ul style="list-style-type: none"><li>Awarded to approximately 20 undergraduate students in Taiwan every semester.</li></ul> | Sept. 2020 - Jan. 2021 |
| <b>The Dean's List Award</b> <ul style="list-style-type: none"><li>Ranked 1st, with 4.30/4.30 GPA in the semester.</li></ul>                                                   | Spring 2020            |